Project - Low Level Design

On

Marketing Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

This document presents Marketing Planning Assistant Agent is an Agentic AI system built to automate the process of creating a complete marketing campaign plan from a single high-level goal. Traditional marketing planning requires a team of specialists — a researcher, a strategist, a scheduler, and a report writer working over several days to produce a campaign document. This project replaces that manual process with five AI agents that execute sequentially and deliver a structured marketing plan in under five minutes.

The project is built using CrewAI as the multi-agent orchestration framework, Groq as the free-tier LLM provider running the llama-3.1-8b-instant model, Python 3.12 as the programming language, and Streamlit for the web interface.

1.1. Scope of the document

This document defines the detailed low-level design of all agent, task, tool, and utility classes , Sequential data flow from user input through five AI agents to final output ,User flow of the Streamlit web application ,Entity definitions and relationships between system components ,Technology stack selection .

1.2. Intended Audience

This document is intended for:

- Developers implementing the system
- Academic evaluators and mentors
- Students studying Agentic AI
- Researchers exploring AI in Marketing Planning Assistant Agents

It helps the audience understand system design, workflows, and technical implementation details.

1.3. System overview

The system is a multi-agent AI-based Marketing Planning assistant that processes user-defined goals and generates structured marketing plans.

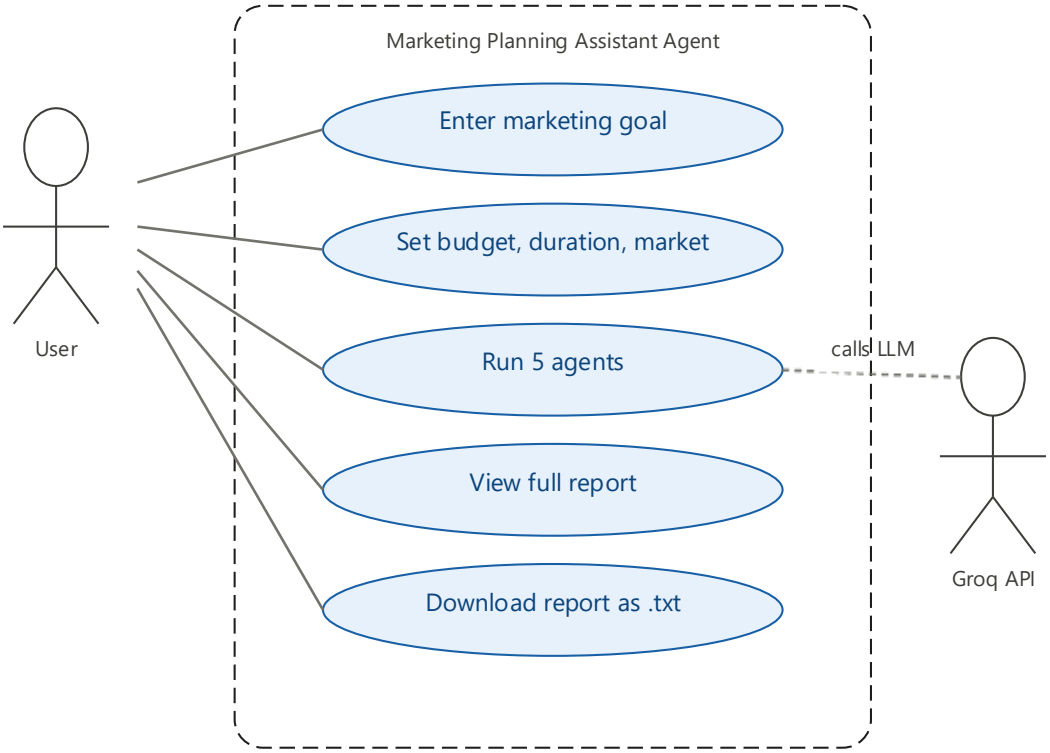

Figure 1: Use Case Diagram

Working Overview

1. User provides a Marketing goal
2. Planner Agent interprets the goal
3. Goal is decomposed into smaller tasks
4. Specialized agents perform tasks
5. Tools validate data and resources
6. Final execution plan and report is generated

Key Features

- Multi-agent collaboration
- Tool-based validation
- Dynamic planning and scheduling

2. System Design

The system follows a modular multi-agent architecture where each component performs a specific role. The design ensures flexibility, scalability, and efficient task execution.

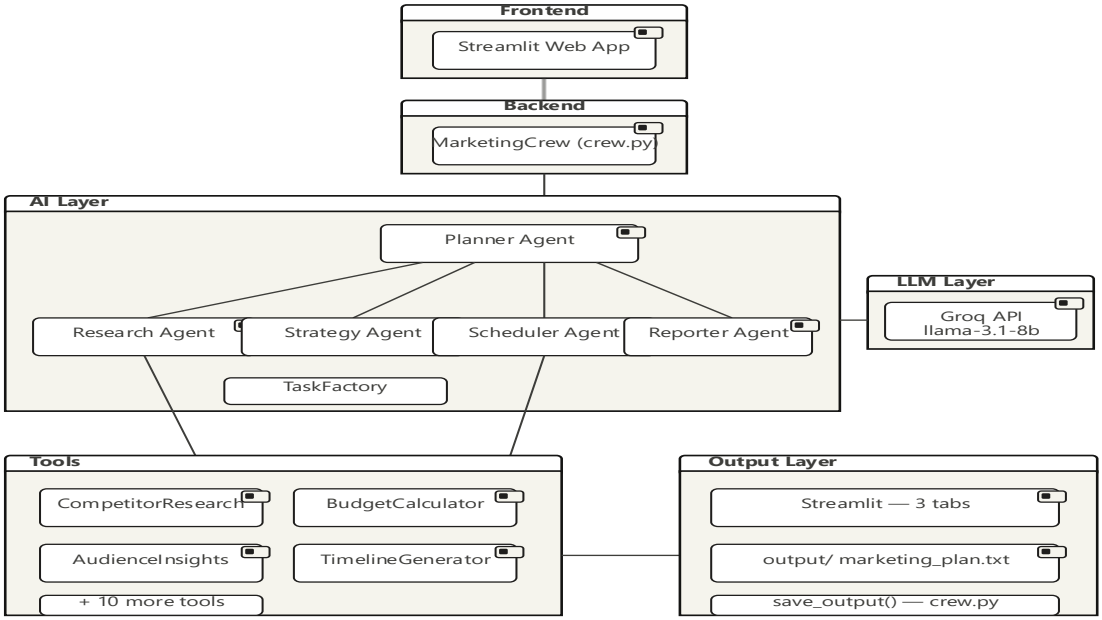

Figure 2: Architecture Diagram

Architecture Flow

User Input → Planner Agent → Task Decomposition → Agents → Tools → Output

2.1. Application Design

The application consists of the following modules:

1. Input Module

- Accepts marketing goal from user in natural language
- Validates that the goal is non-empty and meaningful
- Enriches input with additional parameters — budget, duration, and target market
- Passes the final goal string to the Planner Agent

2. Planner Agent

- Core decision-making unit
- Breaks goals into smaller tasks using LLM

3. Agent Layer

- Research Agent analyzes the market, identifies 3 competitors with their messaging and weaknesses, builds 2 audience personas, and finds 5 relevant market trends
- Strategy Agent builds the campaign strategy including positioning statement, channel mix, budget allocation across channels, and 7 measurable KPIs with targets and deadlines
- Scheduler Agent creates the 12-week execution timeline divided into 4 phases — Pre-Launch, Launch, Optimize, and Evaluate — with task owners and dependencies
- Reporter Agent aggregates all previous agent outputs and writes the complete 8-section executive marketing plan report

4. Tool Layer

- Competitor Research Tool
- Audience Insights Tool
- Market Trends Tool
- Budget Calculator Tool
- Channel Selector Tool
- KPI Generator Tool
- Timeline Generator Tool
- Dependency Mapper Tool
- Resource Allocator Tool
- Report Formatter Tool
- Risk Assessor Tool

5. Output Module

- Aggregates results
- Generates structured Marketing plan

2.2. Process Flow

The system follows a structured execution flow:

1. User inputs marketing goal
2. System validates input
3. Planner Agent decomposes the goal
4. Tasks are assigned to agents
5. Agents execute tasks
6. Tools are used for validation
7. Results are combined
8. Final plan is generated

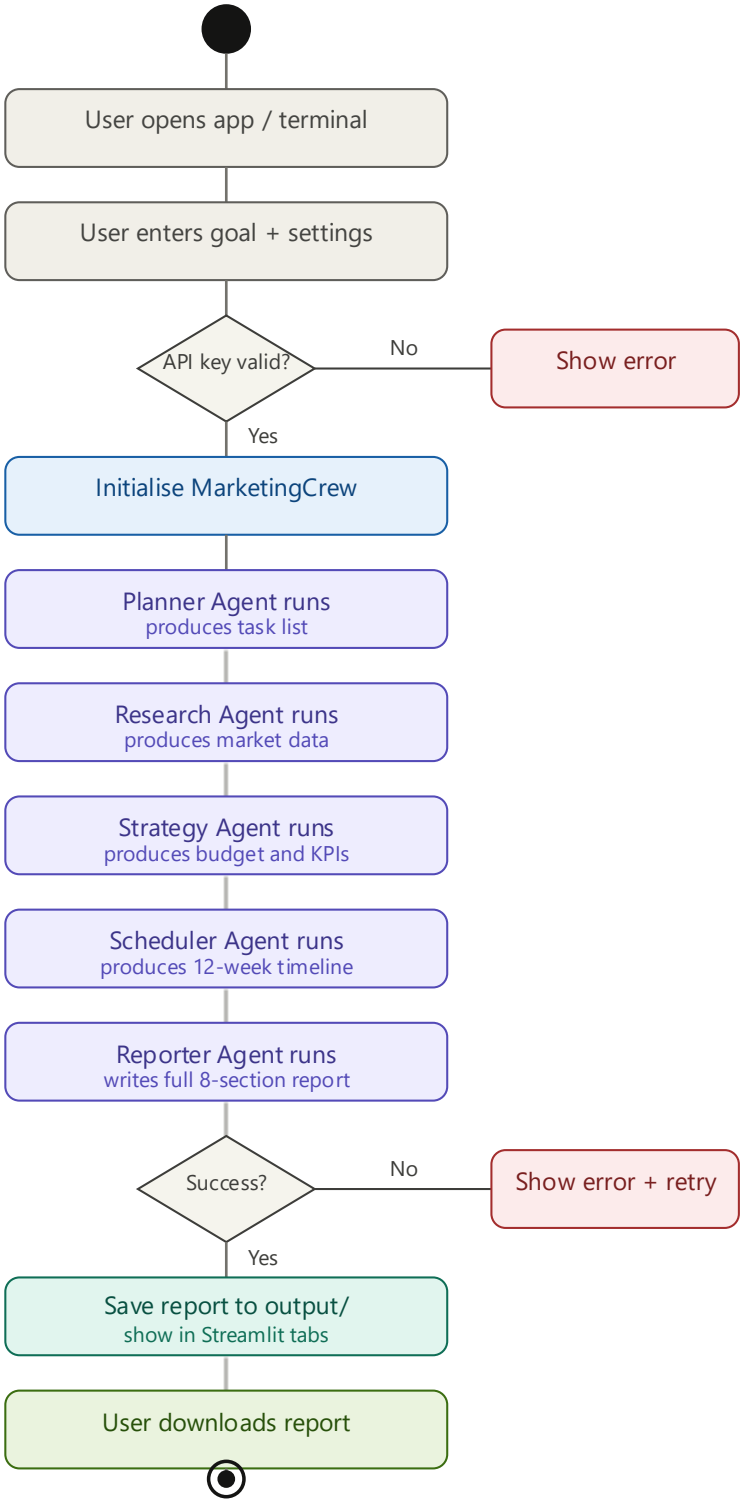

Figure 3: Activity Diagram

2.3. Information Flow

The flow of information in the system is as follows:

User Input
↓
Planner Agent
↓
Task List
↓
Agents
↓
Tool Calls
↓
Processed Data
↓
Final Output

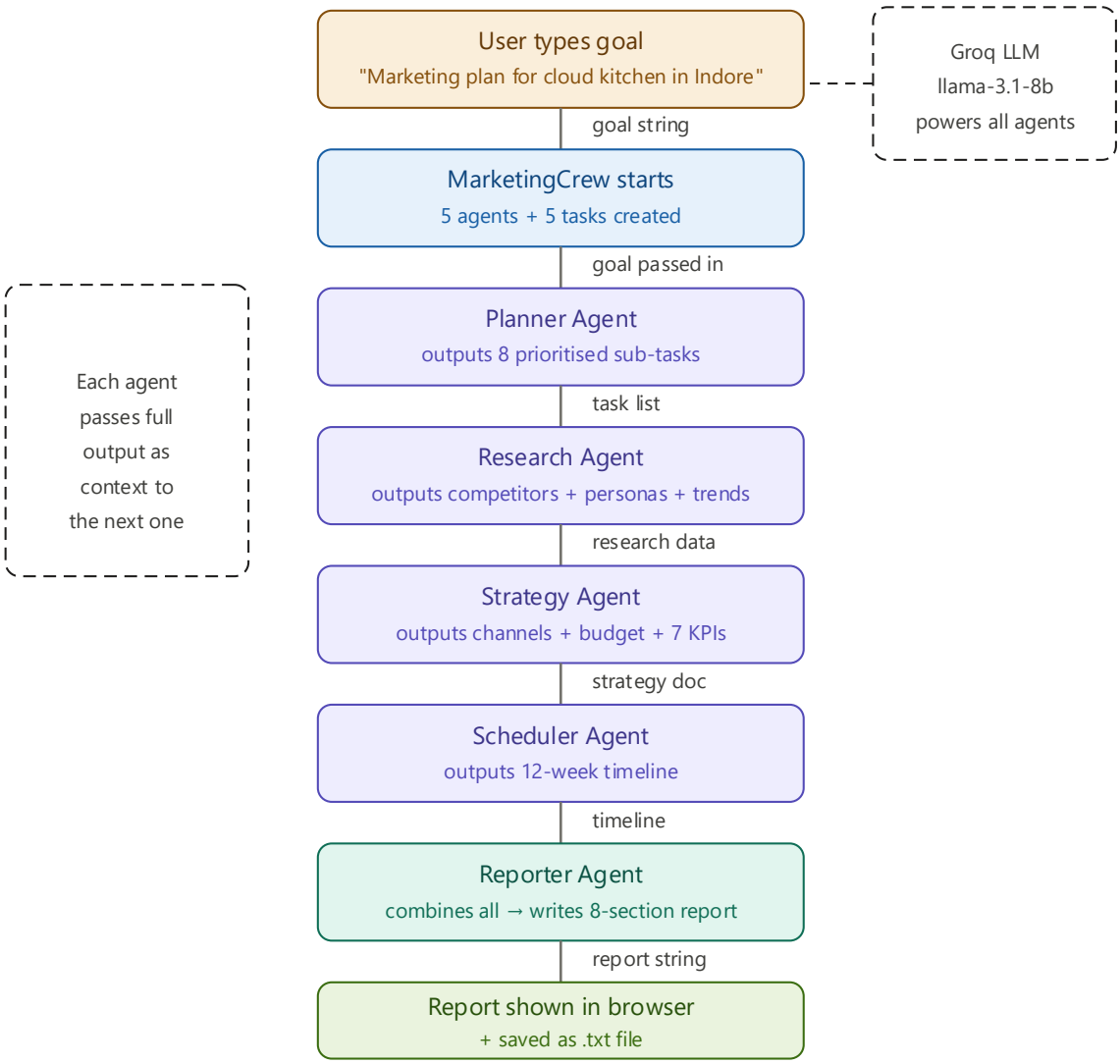

Figure 4: Flow Chart

Data Handling

- Input is processed and structured
- Intermediate data is shared between agents
- Final output is formatted for readability

2.4. Components Design

1. Planner Agent

- Responsible for task decomposition
- Uses LLM for reasoning and planning

2. Research agent

- Analyzes patient symptoms
- Identifies possible conditions

3. Strategy Agent

- Builds the full campaign strategy from research findings
- Selects the top 3 to 4 marketing channels with budget justification

4. Scheduler Agent

- Converts strategy into a 12-week execution timeline
- Divides work into 4 phases — Pre-Launch (Weeks 1–3), Launch (Weeks 4–6), Optimize (Weeks 7–10), Evaluate (Weeks 11–12)
- Assigns owner roles to each task and identifies critical path dependencies

5. Reporter Agent

- Synthesizes all previous agent outputs into a polished executive document
- Writes 8 sections — Executive Summary, Goal, Research, Strategy, Schedule, Risks, Quick Wins, Appendix
- Adds risk assessment with top 3 risks and mitigations

6. LLM Integration

- Uses Groq API
- Model: llama3-8b-8192
- Temperature = 0.3
- Max_token = 600 per agent within rate limits

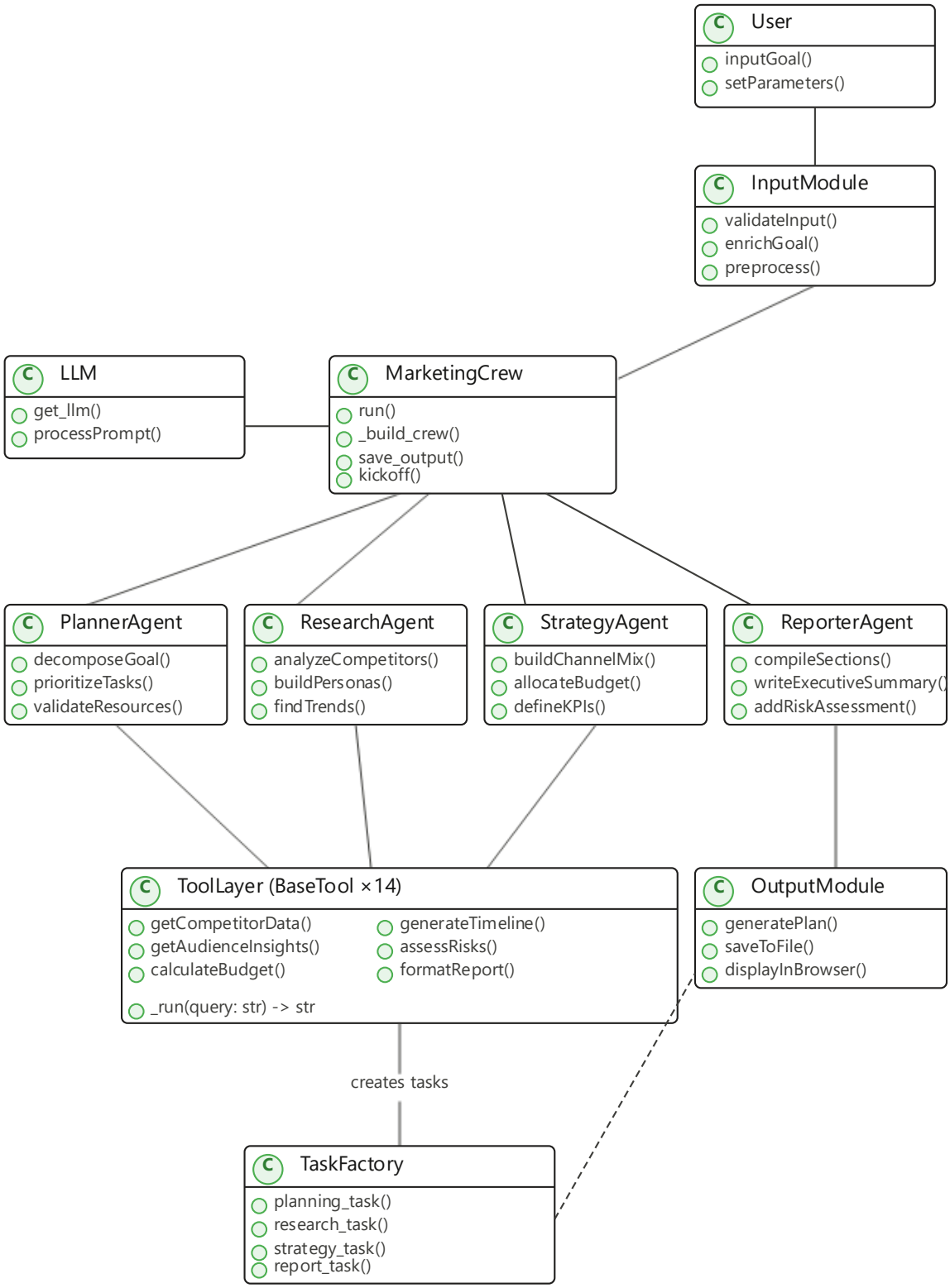
User
inputGoal()
setParameters()
InputModule
validateInput()
enrichGoal()
preprocess()
LLM
get_llm()
processPrompt()
MarketingCrew
run()
_build_crew()
save_output()
kickoff()
PlannerAgent
decomposeGoal()
prioritizeTasks()
validateResources()
ResearchAgent
analyzeCompetitors()
buildPersonas()
findTrends()
StrategyAgent
buildChannelMix()
allocateBudget()
defineKPIs()
ReporterAgent
compileSections()
writeExecutiveSummary()
addRiskAssessment()
ToolLayer (BaseTool ×14)
getCompetitorData()
getAudienceInsights()
calculateBudget()
generateTimeline()
assessRisks()
formatReport()
_run(query: str) -> str
OutputModule
generatePlan()
saveToFile()
displayInBrowser()
creates tasks
TaskFactory
planning_task()
research_task()
strategy_task()
report_task()

Figure 5: Class Diagram

2.5. Key Design Considerations

Scalability

The system can handle multiple agents and tasks simultaneously.

Modularity

Each component is independent and can be updated separately.

Flexibility

New agents and tools can be added easily.

Performance

Fast response using efficient LLM models.

Reliability

Mock tools ensure stable execution without external dependencies.

Security

No real patient data is used, ensuring data privacy.

2.6. API Catalogue

1. LLM API (Groq)

Parameter	Description
API Key	Authentication
Model	llama3-8b-8192
Input	Prompt
Output	Generated response

2. Competitor Research API (Mock)

Input:

Industry or market word

Output:

Competitor profiles with ad spend, channels, messaging, and weakness

3. Audience Insight API (Mock)

Input:

Product or target marketing description

Output:

Competitor profiles with ad spend, channels, messaging, and weakness

4. Budgetcalculator API (Mock)

Input:

Campaign goal type and total budget

Output:

Channel-wise budget allocation with percentage split and rationale

3. References

- CrewAI Documentation
- LangChain Documentation
- Groq API Documentation
- Research papers on Agentic AI
- Marketing Planning Assistant AI system design resources